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Group Details:

	Student ID	Student Name	Topics
1	IT22275738	Dassanayake A D D K K	• Introduction, • Security inspection tools in Linux.
2	IT22156556	Deshapriya D L N S	• Effective techniques for maintaining Security, • Virtual File System.
3	IT22155948	Jayathissa M T N	• Advanced Security Measures, • Appendices.
4	IT22121592	Herath H M C H K	• Core security principles in Linux, • Conclusion.

Security Protection and Inspection Method for Confidential Documents Based on Linux Operating System

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Introduction

In the constantly changing digital world, where cyberattacks can happen at any time, system security is critical. Linux is a robust, open-source operating system that has gained popularity due to its built-in security features and adaptability. But even the strongest stronghold needs ongoing attention. This is where Linux's security features and inspection tools are useful.

This domain includes a wide range of instruments, methods, and best practices intended to strengthen your Linux system against possible weaknesses and increase its resistance to constantly evolving threats. Linux provides a strong security toolkit that includes proactive vulnerability detection via vulnerability scanning and suspicious activity analysis of system logs.

By using these tools wisely and following security guidelines, you can give yourself the ability to:

- Detect emerging threats before they can exploit vulnerabilities.
- Mitigate potential risks by taking timely action to address weaknesses.
- Respond effectively to security incidents, minimizing damage and downtime.

Core Security Principles in Linux

User management in Linux.

Managing users involves overseeing and controlling user access, to resources in a system or application. This includes tasks like creating, changing, and removing user accounts and assigning permissions based on roles.

User management is essential for security, access control, and smooth operations in computer systems or networks. User management systems offer features such as authentication (confirming user identity) authorization (deciding resource access) and auditing (tracking activities, for accountability). User management can be manual or automated using software tools. These tools allow administrators to handle user accounts, permissions, and activity monitoring.

As mentioned above user management, in Linux, includes tasks such as setting up adjusting, and removing user profiles along with controlling permissions to limit or grant entry to files and directories. In the realm of Linux, one encounters user categories, each fulfilling roles and possessing differing levels of system resource access.

Let's delve into the user types:

Standard Users:

These individuals are users engaging with the system, for activities like file access, application execution, and routine computing tasks. Their privileges are restricted compared to superusers (root). Do not extend to performing system administrative functions.

- Standard users typically possess:
    - A personal home directory for storing files and configurations.
    - Limited entry, to system configuration files and critical system resources.

Root Users:

The superuser, commonly known as "root " possesses access, to all system resources and is capable of carrying out any task within the system. Root permissions are essential for actions

that impact the system, such as software installation, configuration adjustments, and user and group management.

- Key attributes of the root user comprise:
    - UID: Root is recognized by the user ID.
    - Total authority, over the system enabling modifications to system files, software installations, and execution of commands.

Sudo Privileges:

Sudo, which stands for "superuser do " is a command line tool that empowers users to run commands as either the root user or another user with privileges.

Users granted privileges can handle tasks without needing direct access as the root user. This approach enhances security by restricting access, to operations.

We can delve deeper into the user management process involving creating, modifying, and removing user accounts as well as managing permission:

User accounts:

Creating User Accounts-

The command ‘useradd ‘is Used to create a user account.

‘sudo useradd username’

Modifying user accounts-

The command ‘usermod’ is used to make changes to an existing user account. We could use the command to change username, and user ID, Change home directory, lock or unlock a user account, and expire or set password expiry date.

‘ usermod [options] username ‘

Deleting user accounts-

The command ‘userdel’ is used to delete a user account.

‘ sudo userdel username ‘ IV. User Password -

The command ‘passwd’ is Used for setting or updating a user's password.

User Groups:

Groups help organize users and define permissions. Each user is part of one group, often their group named after their username, and may also belong to other groups.

Creating groups-

The command ‘groupadd’ is used to create groups.

‘ sudo groupadd groupname ‘

Modifying groups.

The command ‘groupmod’ is used to modify a group. For example, change group name, change group ID, add user to group.

‘ groupmod [options] groupname ‘

Deleting groups.

The command ‘groupdel’ is used to delete groups.

‘ sudo groupdel groupname ‘

Permissions:

File Permissions:

In managing files and directories there are three types of permissions to consider: read (‘r’) and write (‘w’). Execute (‘x’).

These permissions are assigned to three categories of users: the owner, the group, and others.

Changing File Permissions:

To modify permissions you can utilize the chmod command.

chmod u+r file.txt

This action provides read permission, to the file owner.

Ownership:

Every file or directory is linked to an owner and a specific group.

To alter ownership details you can employ the command.

chown username: groupname file.txt

This procedure changes both the owner and group of the file to match a user and group.

4- *Access control* -

- Sudo Access:

Users have the option to receive privileges by editing the /etc/sudoers file or by adding themselves to the sudo group.

This enables them to run commands with permissions using sudo.

File System Permissions:

In Linux systems, there is a default permission structure in place that regulates access to files and directories.

These permissions can be configured accordingly to control access, for users or groups.

III- Service Accessibility:

Many services operate under user accounts and groups to limit entry, to system assets.

Managing service access can involve adjusting setup files or utilizing security measures such, as SELinux or AppArmor.

Virtual File System:

The Virtual File System is an abstraction layer in Linux operating systems that enables to interaction with multiple file systems, and it is a significant feature for security considerations in Linux distributions. Also, it establishes the incorporation of secure file systems, which have certain requirements for various security architectures. [“Appendices A”]

Moreover, Virtual File Systems are encouraged to use encrypted file systems, which further strengthens security. An encrypted file system is a secure technique for users to ensure data confidentiality and defend sensitive data from malicious activities such as unauthorized access.

Virtual file systems facilitate the Access Control mechanism, such as filter file access permission and access control lists to add restrictions to access the files and directories so that it prevents unauthorized access to critical data.

Additionally, Linux Security Modules provide File Integrity Monitoring tools, such as AIDE and Tripwire to monitor unauthorized changes to critical system files. This feature allows for the identification of unauthorized modifications and data breaches that bolster system protection against security threats.

Virtual File Systems enable a variety of security implementations that ensure system and user data protection against security threats. Virtual file system facilitations such as support for various file systems, encrypted file methods, access control, and file integrity monitoring tools are fortifying the security of Linux operating systems.

Security Inspection Tools in Linux

Linux is a stable and secure operating system that stands out in the constantly changing digital landscape. But even the safest systems need ongoing attention to detail. Linux security inspection tools are essential for spotting security flaws, unusual activity, and possible breaches.

With the help of these tools, users can keep a strong security posture and proactively safeguard their systems.

Vulnerability Scanners**:**

These tools automate the process of finding system flaws and incorrect configurations. They contrast the configuration of the system with known flaws in the kernel, libraries, and software.

Examples:

OpenVAS (Open Vulnerability Assessment Scanner): A free and open-source scanner known for its extensive vulnerability database and comprehensive reporting capabilities.

Nessus (Professional and Open-Source Editions): A popular commercial scanner with a vast vulnerability database and advanced features like social engineering testing and compliance reporting (Professional edition).

Packet Sniffers**:**

Network traffic passing through a network interface is captured and examined by packet sniffers.

This makes it possible for experts in network security to spot shady activities, recognize illegal attempts at network access, and resolve problems with network connectivity.

Examples:

Wireshark: A widely used and free open-source packet sniffer with a user-friendly interface and powerful filtering capabilities for in-depth traffic analysis.

tcpdump: A command-line-based packet sniffer included in most Linux distributions. It offers a lightweight and efficient way to capture network traffic for basic analysis.

Password Crackers**:**

These tools attempt to crack passwords using various techniques like dictionary attacks and brute-force methods.

Important Note: Password crackers should only be used ethically on authorized systems with permission, such as testing the strength of user passwords.

Examples:

John the Ripper: A well-known open-source password cracker that utilizes various cracking algorithms and supports password hashes from different systems. (**Use ethically!**)

Hashcat: Another powerful password-cracking tool with advanced features like custom mask generation and GPU acceleration. (**Use ethically!**)

Log Analysis Tools**:**

System logs record various events and activities within a Linux system. Log analysis tools help parse these logs, identify anomalies, and detect potential security incidents.

Examples:

Logwatch: A free and open-source tool that automatically analyzes system logs and generates reports highlighting potential security issues. **Graylog:** A popular open-source log management platform that offers centralized log collection, analysis, and visualization for real-time monitoring.

Security Information and Event Management (SIEM) Tools**:**

SIEM tools combine information from multiple sources, such as network traffic, vulnerability scans, and logs, to elevate security inspection to a new level.

They offer a comprehensive perspective of security events, making threat identification simpler and reaction times quicker.

Examples:

ELK Stack: A free and open-source SIEM solution consisting of Elasticsearch (search engine), Logstash (log collection), and Kibana (data visualization).

Splunk: A commercially available SIEM platform that offers advanced features like threat intelligence, incident response, and user behavior analytics.

Choosing the Right Tool

Your unique needs and level of experience will determine which security inspection tool is best for you.

OpenVAS could be adequate for simple vulnerability scanning. A useful tool for thorough network analysis is Wireshark.

Analysis can be expedited when working with large volumes of logs by using a tool such as Graylog.

For enterprises needing extensive security monitoring and incident response capabilities, SIEM tools are perfect.

Advanced Security Measures

Nowadays, a security team does not know what they will face next with the increasing number of endpoint devices and growing reliance on cloud-based services, the potential attack surface area is expanding. According to those factors, it becomes difficult for security teams to keep track of events happening across an enterprise network.

And also, attackers do not only focus on a single system or software, they launch distributed attacks on multiple systems making it difficult for the existing security measures to detect unusual activity.

What is SIEM(Security Information Event Management)?

So, using SIEM we can easily apply this support to threat detection which can be ruled-based or employ a statistical correlation to make connections between event log entries and it generates security alters when it identifies potential security issues.

This system(SIEM) combines Security Information Management(SIM) and Security Event Management(SEM). SIM primarily looks after data collection from log sources and SEM performs real-time monitoring of enterprise systems for event correlation and threat detection.

How does SIEM work?

Data collection

Data storage

Policies and rules

Data consolidation and correlation

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SIEM system collects data via deploying collection agents on end-user devices, servers, and other security systems (like a firewall) After normalizing the data, it users threat intelligence, inbuilt rules, and advanced analytical functions to detect security incidents in real-time and it offers single pane holistic view of an organization's information security.

When SIEM identifies a threat, it generates alters to notify the security team which based on rules.

Why do we need SIEM?

This SIEM solutions provide methods for threat detection and long-term analytics of security logs and those solutions support the security team in responding to potential security incidents faster. As a result, the team can focus on alters with high risk and significant impact.

This system can also help an organization meet compliance requirements by automatically generating reports that include all the logged security events among this source.

There a some kinds of benefits using SIEM. Such as,

It is taken to identify threats significantly and make it easier to gather and analyze security information to keep safe.

Also, it improves cyber security performances and can be configured to detect activity related to an attack rather than the attack itself.

It has a wide variety of users for anything that revolves around data or historical logs.

Provides threat detection and security alters.

Despite all the benefits, SIEM is not a perfect solution, like all cyber security measures. It has limitations.

Takes a long time to implementation.

It requires at least that amount of time to successfully integrate with an organization's security controls and the hosts inside its infrastructure.

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Expensive

- **SIEM features and capabilities**

Data aggregation

Data is collected from applications, databases

Correlations

This refers to the tool finding similar attributes between different events

Dashboards

Display in charts to help find patterns and avoid missing critical events

Altering

Integrations with other controls

The system can give commands to other enterprise security controls to prevent or stop attacks

AI(Artificial Intelligence)

The system improves its accuracy through machine learning

Forensic capabilities

Can the system capture additional information about security events

- **SIEM tools and software**

These tools provide real-time visibility across an organization's information security systems and automatic security event notifications.

IBM QRadar- security monitoring for IT infrastructure

Exabeam- advanced analytics and threat hunter

Datadog Cloud SIEM- cloud-native network and management system

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Effective Techniques for Maintaining Security in Linux

Safeguarding the Linux operating system is most important to protect from various types of vulnerabilities and threats in cyber security. The Linux operating system is known for its robustness and flexibility, used in different environments, including personal computer systems, IoT devices, and server systems, making it attractive for cybercrime.

However, users must follow effective methods to ensure their protection of user identities, sensitive information, virtual documents, and other user assets from cybercriminals. Ensuring the integrity and continentality of the system may defend user information and prevent unauthorized access.

- File Encryption – The use of file and directory encryption technology tools such as Linux Unfiled Key Setup or GnuPG could be an effective method for protecting sensitive data and information and enhancing data confidentiality. Files and Folders could not be accessed without the decryption key, which would prevent unauthorized access.
- Monitoring System Logs – Monitoring system logs are essential for maintaining the system's protection, performance, and server health. While, using critical log files monitoring, such as generic system activity logs, authentication logs, application logs, and system logs to identify malicious activities within the system and respond to the security incidents in real-time.
- Regular System Updates – Operating system patch updates might include security patches that fix system vulnerabilities and threats. Keeping the system up to date with the latest patches of security is essential for preventing malicious attacks by hackers, and it is protective of user privacy and sensitive information within the Linux system.
- User Authentication – A robust authentication method could prevent unauthorized access to the user account. Users must change their user account passwords regularly to avoid password leaks, enable multi-factor authentication mechanisms to create additional security layers, utilize tools for account lockout policies.
- Secure Network Communications – Virtual Private Network (VPN) establishes the connection as more secure over public networks, implements secure protocols, such as Secure Shell (SSH) for remote access; TLS/SSL protocols to encrypt the network

- traffic; and beware of using Hyper Text Transfer Protocol (HTTP) to minimize the risk of attacks and data interception.
- Use Firewalls – configuring Firewalls, such as iptables, to prevent unauthorized entry and protect private data within local networks. Firewalls Control outgoing and incoming network traffic, limit access privileges and perform traffic filtering to decrease the risk of or prevent attacks including virus attacks and malware that damage the system, causing slow operations or permanent data losses.
- Perform System Backups – System backups provide a safety layer against disaster of data losses. Eliminating the impact of malicious attacks, such as Ransomware attacks, and quick data recovery in the event of system failures are the impotence of regular system backup progress.

Conclusion

The Virtual File System (VFS) is a component, in safeguarding the security of Linux operating systems. It acts as a layer that simplifies interactions with file systems while prioritizing security measures. VFS allows for the integration of file systems tailored to security needs across various architectures.

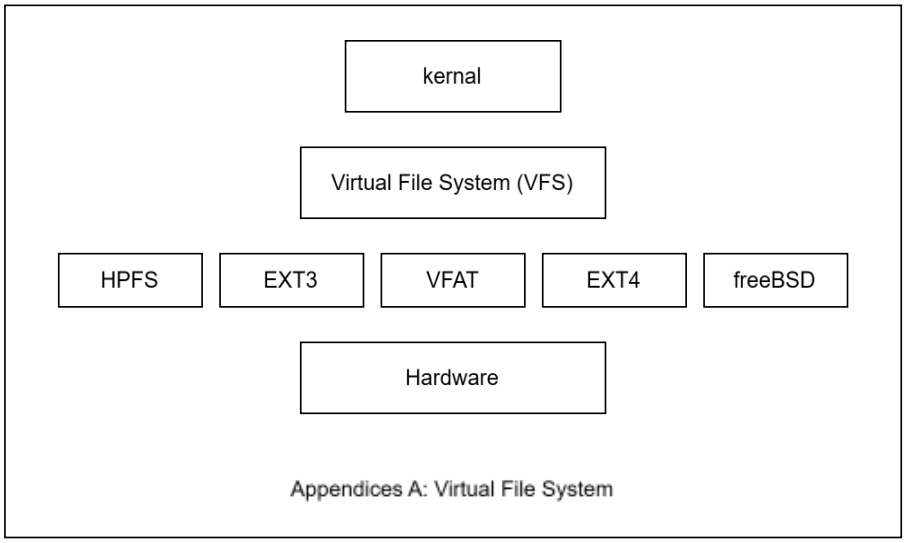
An important feature of VFS is its ability to support file systems, which enhances data confidentiality and shields sensitive information from access. By implementing access control mechanisms like filtering file permissions and access control lists VFS helps thwart entry to data.

Furthermore, Linux Security Modules (LSM) leverage VFS to deploy File Integrity Monitoring (FIM) tools such as AIDE and Tripwire. These tools oversee any illicit alterations made to system files aiding in the detection of modifications and data breaches thus fortifying system defenses against security risks.

In essence, VFS facilitates a range of security measures that safeguard both system and user data from emerging security threats. Its backing for file systems encrypted file techniques, access controls and file integrity monitoring tools reinforces the security stance of Linux operating systems enhancing their resilience, against cyberattacks.

Appendices

- Appendices A: Virtual File System



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